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ATTEMPT 2/2 (SUBMITTED 09/04/2026, 00:11) ▾

RECEIPT: CC7A3506C3E844769A22331A61A25D87

100 / 100

ASSESSMENT CONTENT



1

MULTIPLE CHOICE

CORRECT

3 / 3



What is the primary advantage of Attribute-Based Access Control (ABAC) over Role-Based Access Control (RBAC)?

- ☐ A ABAC is simpler to implement than RBAC
- ☒ B **Correct:** ABAC allows fine-grained control without needing to create many predefined roles Correct answer
- ☐ C ABAC requires fewer resources to operate with fewer roles
- ☐ D ABAC only works in military environments with specific clearances and labels

Feedback

Good work.

ABAC is highly scalable as it allows fine-grained control without needing to create many predefined roles. Instead of assigning roles, ABAC evaluates multiple attributes such as user department, device type, location, and time to determine access permissions dynamically.

Check week 10 notes for more details.

2

MULTIPLE CHOICE

CORRECT

2 / 2



Which CIA triad objective is violated when unauthorised modification of patient information occurs in a hospital system?

☒ A **Correct:** Integrity

Correct answer

☐ B Availability

☐ C Authenticity

☐ D Confidentiality

Feedback

Good work.

Integrity ensures that information and programs are changed only in a specified and authorised manner. Unauthorised modification of data violates integrity, which can result in serious consequences, especially in critical systems like healthcare where inaccurate information could harm patients.

Check week 1 notes for more details

3

MULTIPLE CHOICE CORRECT

2 / 2



Which of the following best describes operational controls?

☒ A **Correct:** Controls implemented by people to address security policies

Correct answer

☐ B Controls that focus on security planning and guidelines

☐ C Controls that involve the use of hardware and software security capabilities

☐ D Controls that provide network encryption

Feedback

Good work.

Operational controls address the correct implementation and use of security policies and standards, ensuring consistency in security operations. These controls relate to mechanisms and procedures primarily implemented by people rather than systems.

Check week 9 notes for more details.

4

MULTIPLE CHOICE CORRECT

1 / 1



Which breach threat involves intercepting information during transmission?

- ☐ A Repudiation
- ☒ B **Correct:** Eavesdropping
- ☐ C Interruption
- ☐ D Alteration

Correct answer

 Feedback**Good work.**

Eavesdropping is passive monitoring of communications without authorisation. It involves intercepting information during transmission to gain unauthorised access to sensitive data.

Check week 1 notes for more details

5

MULTIPLE CHOICE CORRECT

2 / 2



Which access control model assigns security labels such as "Top Secret" or "Confidential" to users and data?

- ☐ A Attribute-Based Access Control (ABAC)
- ☐ B Discretionary Access Control (DAC)
- ☐ C Role Based Access Control (RBAC)
- ☒ D **Correct:** Mandatory Access Control (MAC)

Correct answer

 Feedback**Good work.**

In MAC, the system assigns security labels to users and data. Users can only access data if their clearance level matches the data classification. This model is often used in military and government systems.

Check week 10 notes for more details

6

MULTIPLE CHOICE CORRECT

1 / 1



At which TCP/IP layer does IPsec operate?

- ☒ A **Correct:** Network layer
- ☐ B Transport layer
- ☐ C Physical layer
- ☐ D Application layer

Correct answer

 Feedback

Good work.

IPsec operates at the network layer of the TCP/IP stack, providing security for IP traffic by encrypting and authenticating data at the IP level.

Check week 9 notes for more details.

7

MULTIPLE CHOICE CORRECT

2 / 2



In the DREAD model, what does the 'E' stand for?

- ☐ A Evaluation
- ☐ B Enumeration
- ☒ C **Correct:** Exploitability
- ☐ D Encryption

Correct answer

 Feedback

Good work.

In DREAD, Exploitability measures how easily a vulnerability can be exploited by an attacker. This category helps quantify the technical difficulty required to successfully leverage a particular weakness in the system.

Check week 2 notes for more details

8

MULTIPLE CHOICE CORRECT

1 / 1



What does CSRF stand for?

- ☐ A Cross-System Request Forgery
- ☒ B **Correct:** Cross-Site Request Forgery
- ☐ C Cross-Site Response Failure
- ☐ D Cross-Server Request Failure

Correct answer

 Feedback

Good work.

CSRF stands for Cross-Site Request Forgery. It is a web security vulnerability that allows an attacker to induce a user to perform actions without their consent by exploiting the trust a website has in the user's browser.

Check week 7 notes for more details.

9

FILL IN THE BLANK CORRECT

5 / 5



Decrypt the ciphertext "FWF STJXQJWX" using the provided Vigenère Polyalphabetic Substitution table and the keystream "SIMPLE". The plaintext is **✓ NOT DIFFICULT**

Vigenère cipher table		Keystream																									
		a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Plaintext	a	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
	b	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
	c	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
	d	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
	e	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
	f	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
	g	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
	h	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
	i	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
	j	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
	k	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
	l	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
	m	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
	n	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
	o	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
	p	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
	q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
	r	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
	s	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
	t	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
	u	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
	v	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
	w	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
	x	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
	y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
	z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Correct

Acceptable answers for **BLANK-1** are **NOT DIFFICULT, not difficult, Not Difficult, Not difficult, not Difficult, NOTDIFFICULT, notdifficult, NotDifficult, Notdifficult** and **notDifficult**

Responses must match exactly

Feedback

This is correct, well done. It is essential to remember that when decrypting, for each character, you start from the keystream, and you go down in the column of each respective keystream until you find the encrypted character. You then look at the top of the row to find the plaintext character that corresponds to it.

10

FILL IN THE BLANK

CORRECT

5 / 5



Decrypt the message "Kssh Pygo mr MGX" using the Caesar cipher substitution method and the rotation value of "4". The plaintext is **Good Luck in ICT**

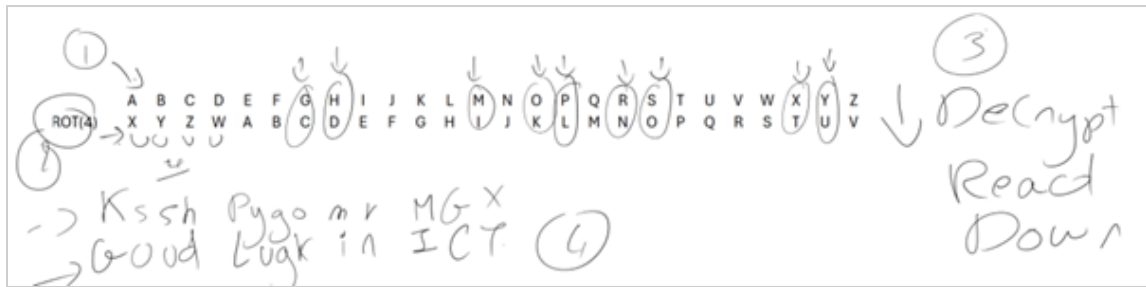
Correct

Acceptable answers for **BLANK-1** are **Good Luck in ICT, good luck in ict, Good Luck In ICT, Good luck in ICT, good Luck In ICT, GOODLUCKINICT, goodluckinict, GoodLuckInICT, Goodluckinict** and **goodLuckInICT**

Responses must match exactly

Feedback

Correct answer: Good Luck in ICT



11

FILL IN THE BLANK

CORRECT

5/5



Decrypt the ciphertext message "AS AEO REWLG TE TUREDCOA" using the transposition method and the secret key "LEARNT". The plaintext is **WE ARE CLOSE TO GRADUATE**

Correct

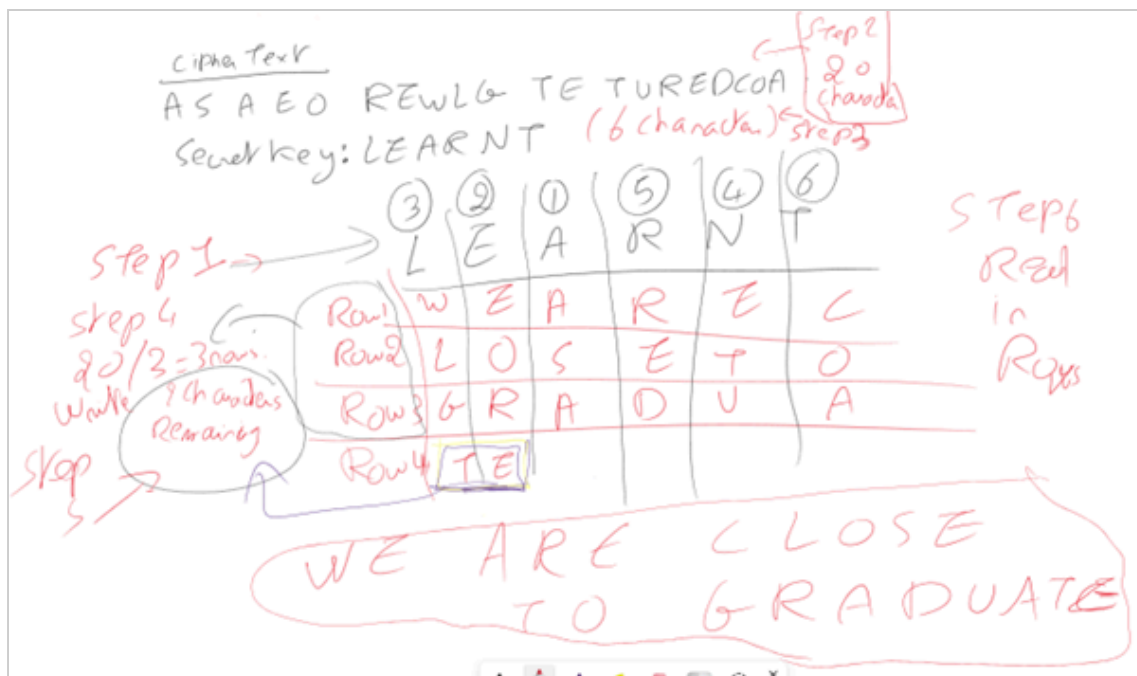
Acceptable answers for **BLANK-1** are **WE ARE CLOSE TO GRADUATE**, **we are close to graduate**, **We Are Close To Graduate**, **We are close to graduate**, **we Are Close To Graduate**, **WEARECLOSETOGRADUATE**, **weareclosetograduate**, **WeAreCloseToGraduate**, **Weareclosetograduate** and **weAreCloseToGraduate**

Responses must match exactly

Feedback

This is correct. **WE ARE CLOSE TO GRADUATE** is correct.

- Step 1- Put a number at the top of each character from the secret work by alphabetical order. If two are the same, then the one that comes first, takes the smaller number.
- Step 2- Count how many characters the ciphertext message is (without spaces). In this example, the ciphertext is composed of **20 characters**
- Step 3- Count how many characters in the secret key. In this example LEARNT is **6 characters**
- Step 4- Divide **20 by 6** will give **3 complete rows** and **2 characters in row 4**
- Step 5- Write the ciphertext in Columns by the alphabetical order set in step 1. Make sure you stop at row 3 in columns with 3 rows and at row 4 in columns with 4 rows (only 2 characters)
- Step 6- READ the plaintext in rows



12

FILL IN THE BLANK

CORRECT

5/5



Encrypt the plaintext message "SUCCEED IN FIRST ATTEMPT" using the transposition method and the secret key "HELLO". The ciphertext is **UDRTSEI TT CISEC NTMEFAP**

Correct

Acceptable answers for **BLANK-1** are **UDRTSEI TT CISEC NTMEFAP**, **udrtsei tt cisec ntmeafap**, **Udrtsei Tt Cisec Ntmeafap**, **Udrtsei tt cisec ntmeafap**, **udrtsei Tt Cisec Ntmeafap**, **UDRTSEITTCISECNTMEFAP**, **udrtseitttcisecntmeafap**, **UdrtseittTcisecNtmeafap**, **Udrtseitttcisecntmeafap** and **udrtseiTtTcisecNtmeafap**
Responses must match exactly

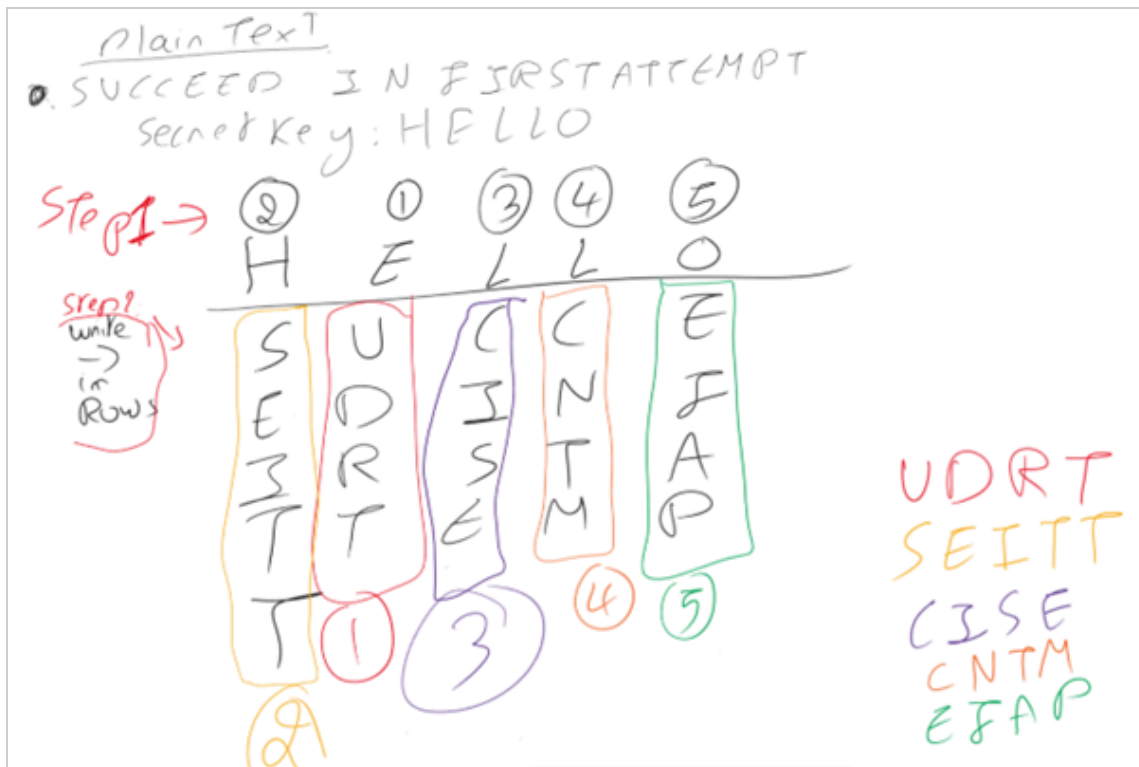
Feedback

UDRTSEI TT CISEC NTMEFAP is the correct answer

Step 1- Put a number at the top of each character from the secret work by alphabetical order. If two are the same, for example L, then the one that comes first, takes the smaller number.

Step 2- Write the plaintext in rows

Step 3- READ the cipher text in columns by the alphabetical order set in Step 1



13

MULTIPLE CHOICE

CORRECT

1 / 1



HTTP basic authentication is insecure because credentials are:

- ☐ A Stored in the server's cache
- ☐ B Encrypted with weak algorithms
- ☒ C **Correct:** Sent in clear text over the network
- ☐ D Protected only by firewall rules

Correct answer

Feedback

Good work.

HTTP basic authentication sends credentials in clear text over the network, making them vulnerable to interception through packet sniffing. The client passes authentication information to the server in an Authorization header without encryption. Check week 3 notes for more details.

14

MULTIPLE CHOICE

CORRECT

1 / 1



What is the primary purpose of a security control?

- ☒ A **Correct:** To reduce risk by eliminating or preventing security violations Correct answer
- ☐ B To increase system performance
- ☐ C To monitor network bandwidth
- ☐ D To manage user accounts

 Feedback

Good work.

A security control is an action, device, procedure, or other measure that reduces risk by eliminating or preventing security violations, minimising harm, or enabling corrective action.

Check week 9 notes for more details

15

MULTIPLE CHOICE

CORRECT

1 / 1



Which type of malware requires a host program to run?

- ☐ A Independent malware
- ☐ B Standalone malware
- ☐ C Transient malware
- ☒ D **Correct:** Host dependent malware Correct answer

 Feedback

Good work.

Host dependent malware requires a host program to run and cannot run independently. It infects a program by inserting its instructions into the program or modifying the host code, unlike independent malware which is a complete program that can run on its own.

Check week 8 notes for more details.

16

MULTIPLE CHOICE CORRECT

2 / 2



Which SQL injection technique adds additional queries beyond the intended query?

☒ A **Correct:** Piggybacked queries

Correct answer

☐ B Inferential attack

☐ C Tautology

☐ D End-of-line comment

Feedback

Good work.

Piggybacked queries involve the attacker adding additional queries beyond the intended query, piggy-backing the attack on top of a legitimate request. This allows multiple SQL commands to be executed in a single request.

Check week 7 notes for more details.

17

MULTIPLE CHOICE CORRECT

3 / 3



In authenticated encryption, which cryptographic goals are achieved_____.

☐ A Integrity and authentication only

☒ B **Correct:** Confidentiality, authentication, and integrity

Correct answer

☐ C Non-repudiation only

☐ D Confidentiality only

 Feedback

Good work.

Authenticated encryption combines both HMAC and CMAC approaches. By encrypting the message and its hash using a symmetric key and a hash function, it ensures confidentiality, authentication, and integrity simultaneously.

Check week 4 notes for more details.

18

MULTIPLE CHOICE **CORRECT**

2 / 2



What is the main disadvantage of an application proxy firewall?

- ☐ A It cannot filter incoming traffic
- ☐ B It cannot authenticate any users
- ☒ C **Correct:** It has additional processing overhead
- ☐ D It is less secure than packet filters

Correct answer

 Feedback

Good work.

Application proxy firewalls must have proxy code for each application and may restrict application features supported. The main disadvantage is the additional processing overhead on each connection.

Check week 11 notes for more details.

19

MULTIPLE CHOICE **CORRECT**

1 / 1



Which of the following is an example of an intangible asset for an organisation?

- ☒ A **Correct:** Customer data
- ☐ B Physical security equipment
- ☐ C IT infrastructure

Correct answer

- ☐ D Network servers

 Feedback

Good work.

Customer data is an intangible asset that organisations must protect. Unlike physical hardware or infrastructure, intangible assets include information such as intellectual property, financial data, and reputation that have value but cannot be physically touched.

Check week 1 notes for more details

20

MULTIPLE CHOICE CORRECT

1 / 1



A TCP SYN scan is also known as:

- ☐ A Full open scan
- ☒ B **Correct:** Half open scan
- ☐ C Stealth scan
- ☐ D Connect scan

Correct answer

 Feedback

Good work.

A SYN scan is also called a half open scan because it sends SYN requests to gather information about open ports without completing the TCP handshake. When an open port is identified, the connection is reset before completion.

Check week 3 notes for more details.

21

MULTIPLE CHOICE CORRECT

2 / 2



In asymmetric encryption, to provide confidentiality, Alice would encrypt the message using_____.

- ☐ A Her own public key

- ☐ B Bob's private key
- ☐ C Her own private key
- ☒ D **Correct:** Bob's public key

Correct answer

Feedback

Good work.

For confidentiality in asymmetric encryption, Alice would encrypt the file with the receiver's public key (Bob's public key). This creates a secure message format that can only be decrypted by the person who has the corresponding private key (Bob).

Check week 4 notes for more details.

22

MULTIPLE CHOICE CORRECT

1 / 1



What is the first stage in the modern hacking methodology?

- ☐ A Gaining Access
- ☐ B Covering Tracks
- ☒ C **Correct:** Reconnaissance
- ☐ D Scanning

Correct answer

Feedback

Good work.

Reconnaissance is the initial stage where attackers passively gather information about their target using public records and Open Source Intelligence (OSINT). This stage helps identify network layouts, domains, servers, and infrastructure details before any active engagement occurs.

Check week 2 notes for more details

23

MULTIPLE CHOICE CORRECT

2 / 2



In Cipher Block Chaining (CBC) mode, what is combined with the current plaintext block before encryption?

- ☐ A The encryption key
- ☐ B The next plaintext block
- ☐ C The initialisation vector only
- ☒ D **Correct:** The previous ciphertext block

Correct answer

Feedback

Good work.

In CBC mode, the previous ciphertext block is XORed with the current plaintext block before encryption. For the first block, an initialisation vector (IV) is used instead of a previous ciphertext block.

Check week 5 notes for more details.

24

MULTIPLE CHOICE CORRECT

2 / 2



In the STRIDE threat model, what does the 'T' stand for?

- ☐ A Tracking
- ☒ B **Correct:** Tampering
- ☐ C Tracing
- ☐ D Transmission

Correct answer

Feedback

Good work.

In STRIDE, Tampering refers to the unauthorised modification of data or system components. This threat category helps identify where attackers might alter information or system behaviour without proper authorisation.

Check week 2 notes for more details

25

MULTIPLE CHOICE

CORRECT

2 / 2



In a session hijacking attack, the attacker must successfully:

- ☐ A Decrypt the session key
- ☐ B Disable the firewall
- ☐ C Install malware on the server
- ☒ D **Correct:** Predict or determine the sequence numbers Correct answer

Feedback

Good work.

For a session hijack to be successful, the attacker must successfully determine or guess the sequence numbers to hijack the session. This allows the attacker to inject packets that appear legitimate and take over the communication.

Check week 3 notes for more details.

26

MULTIPLE CHOICE

CORRECT

1 / 1



Which authentication type uses something you own?

- ☐ A Authentication by location
- ☐ B Authentication by knowledge
- ☐ C Authentication by characteristics
- ☒ D **Correct:** Authentication by ownership Correct answer

Feedback

Good work.

Authentication by ownership uses something you own, such as synchronous tokens, asynchronous tokens, USB tokens, or smart cards.

27

MULTIPLE CHOICE CORRECT

1 / 1



What is the primary purpose of a firewall?

- ☒ A **Correct:** To establish a controlled link between networks Correct answer
- ☐ B To improve network speed and performance
- ☐ C To manage and authenticate user accounts
- ☐ D To store and archive network data

 Feedback

Good work.

A firewall is placed between the premises network and the Internet to establish a controlled link and is used as a perimeter defence to impose security and auditing whilst separating internal systems from external networks.

Check week 11 notes for more details.

28

MULTIPLE CHOICE CORRECT

1 / 1



Which cryptographic goal ensures that data is not altered without authorisation_____.

- ☒ A **Correct:** Integrity Correct answer
- ☐ B Confidentiality
- ☐ C Authentication
- ☐ D Non-repudiation

 Feedback

Good work.

Integrity ensures that data is not altered without authorisation. It verifies that the message received is identical to the sent message and that stored data was not altered between creation and access.

Check week 4 notes for more details.

29

MULTIPLE CHOICE CORRECT

2 / 2



Which worm type exploits an unknown vulnerability that is only discovered when the worm is launched?

- ☒ A **Correct:** Zero-day exploit worm
- ☐ B Multiplatform worm
- ☐ C Multi-exploit worm
- ☐ D Polymorphic worm

Correct answer

Feedback

Good work.

A zero-day exploit worm exploits an unknown vulnerability that is only discovered by the general network community when the worm is launched. This achieves maximum surprise and distribution, as there are no existing defences against the vulnerability. Check week 8 notes for more details.

30

MULTIPLE CHOICE CORRECT

2 / 2



When a UDP port is open, the typical response to a UDP probe is:

- ☐ A RST packet
- ☒ B **Correct:** No response
- ☐ C SYN/ACK packet
- ☐ D ICMP Type 3 message

Correct answer

Feedback

Good work.

No response for a UDP request indicates that the port is open. However, it can also mean that a firewall silently blocked the request. An ICMP Type 3 response indicates a closed UDP port.

Check week 3 notes for more details.

31

MULTIPLE CHOICE

CORRECT

1 / 1



In symmetric encryption, what type of key is used for both encryption and decryption?

- ☐ A A public key
- ☐ B An initialisation vector
- ☒ C **Correct:** A shared secret key
- ☐ D A private key

Correct answer

Feedback

Good work.

In symmetric encryption, both parties use the same shared secret key for encryption and decryption. This key must be kept confidential between the communicating parties.

Check week 5 notes for more details.

32

MULTIPLE CHOICE

CORRECT

1 / 1



What is the primary advantage of asymmetric encryption over symmetric encryption?

- ☐ A It is faster
- ☐ B It uses simpler algorithms
- ☒ C **Correct:** It eliminates the need to securely share a secret key beforehand

Correct answer

- ☐ D It produces shorter ciphertext

 Feedback

Good work.

Asymmetric encryption uses a public and private key pair, eliminating the need for both parties to share a secret key in advance. The public key can be distributed openly whilst the private key remains secret.

Check week 5 notes for more details.

33

MULTIPLE CHOICE CORRECT

2 / 2



Steganography differs from encryption in that it_____.

- ☐ A Provides better confidentiality
- ☐ B Uses stronger algorithms
- ☒ C **Correct:** Hides the existence of data rather than scrambling it Correct answer
- ☐ D Requires larger key sizes

 Feedback

Good work.

Steganography hides data inside other media so its existence is concealed, providing security through obscurity. The message is hidden, not encrypted. Only the sender and receiver know the message exists.

Check week 4 notes for more details.

34

MULTIPLE CHOICE CORRECT

1 / 1



Cryptography is defined as a method of_____.

- ☐ A Breaking encrypted messages

- ☐ B Analysing cipher patterns
- ☒ C **Correct:** Storing and transmitting data in a form that only intended recipients can read and process Correct answer
- ☐ D Hiding data inside other media

Feedback

Good work.

Cryptography is a method of storing and transmitting data in a form that only those it is intended for can read and process. This ensures secure communication and data protection.

Check week 4 notes for more details.

35 MULTIPLE CHOICE CORRECT

1 / 1



What is the primary purpose of a honeypot?

- ☐ A To improve overall network performance
- ☐ B To manage authentication and authorisation
- ☐ C To store backup and archived data
- ☒ D **Correct:** To lure attackers and collect intelligence Correct answer

Feedback

Good work.

Honeypots are decoy systems designed to lure potential attackers away from critical systems, collect information about the attacker's activity, and encourage the attacker to stay on the system long enough for administrators to respond.

Check week 11 notes for more details.

36 MULTIPLE CHOICE CORRECT

2 / 2



What is the key length used by DES?

☒ A **Correct:** 56 bits

Correct answer

☐ B 128 bits

☐ C 64 bits

☐ D 256 bits

Feedback

Good work.

DES uses a 56-bit key length. Although the key is 64 bits in total, 8 bits are used for parity checking, leaving 56 bits for the actual encryption key.
Check week 5 notes for more details.

37

MULTIPLE CHOICE

CORRECT

3 / 3



In symmetric cryptography, if 100 people were going to communicate, the number of keys needed would be_____.

☐ A 200

☒ B **Correct:** 4950

Correct answer

☐ C 9900

☐ D 100

Feedback

Good work.

The equation used to calculate the number of symmetric keys needed is $N * (N - 1) / 2$. For 100 people, this equals $100 * 99 / 2 = 4950$ keys. This demonstrates the scalability challenge of symmetric encryption.
Check week 4 notes for more details.



What is the main difference between a closed policy and an open policy in access control?

- ☒ A **Correct:** Closed policy denies all access by default whilst open policy allows all access by default Correct answer
- ☐ B Closed policy allows all access by default whilst open policy denies all access by default
- ☐ C Both policies work identically
- ☐ D Closed policy requires encryption whilst open policy does not

 Feedback

Good work.

In a closed policy, access is denied by default unless explicitly allowed. In an open policy, access is allowed by default unless explicitly denied.

Check week 10 notes for more details.



What is the primary difference between a vulnerability assessment and a penetration test?

- ☐ A Vulnerability assessments are automated, penetration tests are manual
- ☐ B Vulnerability assessments focus on networks, penetration tests focus on applications
- ☐ C Vulnerability assessments are cheaper, penetration tests are more expensive
- ☒ D **Correct:** Vulnerability assessments scan for weaknesses, penetration tests actively exploit them Correct answer

 Feedback

Good work.

Vulnerability assessments identify and report potential security weaknesses without exploiting them. Penetration tests go further by actively seeking to exploit discovered vulnerabilities to demonstrate real-world impact and validate their severity.

40

MULTIPLE CHOICE

CORRECT

1 / 1



Which malware type actively seeks out more machines to infect using software vulnerabilities?

- ☐ A Virus
- ☐ B Trojan horse
- ☒ C **Correct:** Worm
- ☐ D Rootkit

Correct answer

 Feedback

Good work.

A worm is a program that actively seeks out more machines to infect, with each infected machine serving as an automated launching pad for attacks on other machines. Worms exploit software vulnerabilities in client or server programs to propagate.

Check week 8 notes for more details.

41

MULTIPLE CHOICE

CORRECT

3 / 3



In the PGP encryption process, what is the purpose of encrypting the random symmetric key with the recipient's public key?

- ☐ A To authenticate the sender's identity
- ☐ B To compress the message data
- ☒ C **Correct:** To ensure only the intended recipient can decrypt the symmetric key
- ☐ D To generate a digital signature

Correct answer

 Feedback

Good work.

In PGP, the random symmetric key used to encrypt the message is itself encrypted with the recipient's public key. This ensures that only the recipient, who possesses the corresponding private key, can decrypt the symmetric key and then use it to decrypt the actual message.

Check week 9 notes for more details.

42

MULTIPLE CHOICE

CORRECT

1 / 1



Which CIA triad principle ensures that only authorised subjects can access protected data?

- ☐ A Accountability
- ☒ B **Correct:** Confidentiality
- ☐ C Integrity
- ☐ D Availability

Correct answer

 Feedback

Good work.

Confidentiality ensures that private information is not made available or disclosed to unauthorised individuals. It protects data from being accessed by those who should not have permission to view it.

Check week 1 notes for more details

43

MULTIPLE CHOICE

CORRECT

1 / 1



Which type of hacker uses tools created by others and has limited technical knowledge?

- ☒ A **Correct:** Script Kiddie
- ☐ B Hacktivist

Correct answer

- ☐ C State-sponsored
- ☐ D Ethical Hacker

Feedback

Good work.

Script Kiddies are hackers who use other people's tools and have limited knowledge. They often perform attacks without actually understanding them, relying entirely on pre-built scripts and software from more experienced hackers. Check week 2 notes for more details

44

MULTIPLE CHOICE CORRECT

2 / 2



Which virus concealment strategy mutates with every infection?

- ☒ A **Correct:** Polymorphic virus
- ☐ B Encrypted virus
- ☐ C Metamorphic virus
- ☐ D Stealth virus

Correct answer

Feedback

Good work.

A polymorphic virus mutates with every infection, changing its appearance to evade detection. This is different from a metamorphic virus which completely rewrites itself at each iteration and may change both behaviour and appearance. Check week 8 notes for more details.

45

MULTIPLE CHOICE CORRECT

2 / 2



Which type of cross-site scripting attack stores the malicious script on the server?

- ☐ A Reflected XSS

- ☐ B DOM-based XSS
- ☐ C Session XSS
- ☒ D **Correct:** Persistent XSS

Correct answer

Feedback

Good work.

Persistent cross-site scripting stores the malicious script on the server, which is then automatically executed whenever users visit the affected page. This differs from

